

Overview in General

This file contains a general overview of the data in the graph including node labels and relationships types.

References

- [jqassistant](#)
- [Neo4j Python Driver](#)

Node Labels

Table 1a - Highest node count by label combination

Lists the 30 label combinations with the highest number of nodes. The labels with the lowest node count are listed in table 1b. The total list would sum up to the total number of labels (100%).

The whole table can be found in the CSV report `Node_label_combination_count`.

	nodeLabels	nodesWithThatLabels	nodesWithThatLabelsPercent
0	[Git, Update, Change]	139612	46.013091
1	[Git, Change, Create]	45325	14.938138
2	[Git, Change, Delete]	22795	7.512738
3	[Java, ByteCode, Method, Member]	13778	4.540930
4	[Java, ByteCode, Parameter]	13445	4.431181
5	[Git, Commit]	11562	3.810585
6	[File, Git]	8331	2.745717
7	[Java, ByteCode, ParameterizedType, Bound]	7316	2.411195
8	[Java, ByteCode, Bound]	7304	2.407240
9	[Git, Change, Rename]	7018	2.312981
10	[Java, ByteCode, Member, Field]	3692	1.216803
11	[Java, Value, ByteCode, Annotation]	2955	0.973904
12	[Java, ByteCode, Bound, WildcardType]	2950	0.972256
13	[Xml, Element]	2162	0.712548
14	[Java, ByteCode, Constructor, Method, Member]	2157	0.710900
15	[Xml, Text]	1450	0.477889
16	[Java, ByteCode, Bound, TypeVariable]	1112	0.366491
17	[Java, ByteCode, Method, Member, Lambda]	986	0.324964
18	[Type, File, Java, ByteCode, ResolvedDuplicate...	903	0.297609
19	[Type, File, Java, Class, ByteCode]	868	0.286074
20	[Json, Key]	739	0.243558
21	[Value, Json, Scalar]	723	0.238285
22	[Java, Value, ByteCode, Primitive]	680	0.224113
23	[Type, File, Java, ByteCode, JavaType]	659	0.217192
24	[Java, ByteCode, Method, Member, GenericDeclar...	579	0.190826
25	[Type, File, Java, ByteCode, ExternalType]	419	0.138093
26	[Git, Change, Copy]	298	0.098214
27	[Author, Git, Person]	297	0.097885
28	[Value, Array]	293	0.096566
29	[Committer, Git, Person]	242	0.079758

Chart 1a - Highest node count by label combination

Values under 0.5% will be grouped into "others" to get a cleaner plot. The group "others" is then broken down in Chart 1b.

<Figure size 640x480 with 0 Axes>

Nodes per label combination (more than 0.5% overall)

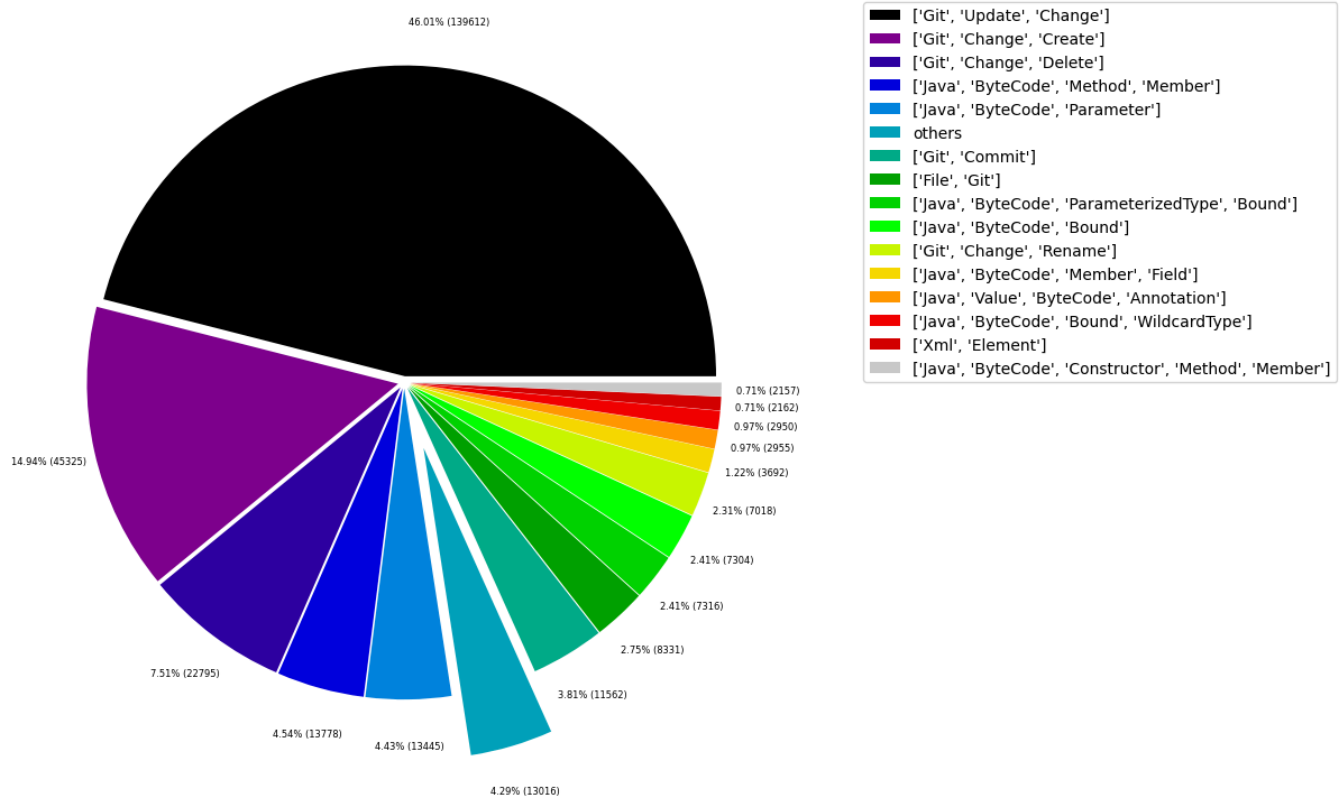


Table 1b - Lowest node count by label combination

Lists the 30 label combinations with the lowest number of nodes until they reach 0.5% of the total node count, which are shown above.

	nodeLabels	nodesWithThatLabels	nodesWithThatLabelsPercent
0	[Analyze, Task, JQAssistant]	1	0.000330
1	[Git, Branch]	1	0.000330
2	[Repository, File, Git]	1	0.000330
3	[File, Json]	2	0.000659
4	[File]	3	0.000989
5	[Java, ByteCode, Constructor, Method, Member, ...]	4	0.001318
6	[Maven, Exclusion]	5	0.001648
7	[Value, Array, Json]	6	0.001977
8	[File, Maven, Xml, Pom, Document]	9	0.002966
9	[Type, File, Java, ByteCode, Void]	9	0.002966
10	[Java, ManifestSection]	9	0.002966
11	[File, Artifact, Jar, Archive, Zip, Java]	9	0.002966
12	[File, Java, Manifest]	9	0.002966
13	[File, Java, ServiceLoader]	11	0.003625
14	[File, Java, Properties]	12	0.003955
15	[Maven, ExecutionGoal]	16	0.005273
16	[Maven, PluginExecution]	16	0.005273
17	[Xml, Attribute]	18	0.005932
18	[JQAssistant, Rule, Concept]	19	0.006262
19	[Type, File, Java, ByteCode, Throwable, Extern...	21	0.006921
20	[Maven, Plugin]	21	0.006921
21	[Maven, Configuration]	21	0.006921
22	[Type, File, Java, ByteCode, Throwable, Resolv...	22	0.007251
23	[Type, File, Java, ByteCode, Enum]	30	0.009887
24	[Type, File, Java, ByteCode, PrimitiveType]	31	0.010217
25	[Xml, Namespace]	36	0.011865
26	[Type, File, Java, ByteCode, Annotation]	44	0.014501
27	[File, Directory]	45	0.014831
28	[Type, File, Java, Class, ByteCode, Throwable]	56	0.018456
29	[Type, File, Java, ByteCode, Interface, Generi...	85	0.028014

Chart 1b - Lowest node count by label combination

Shows the lowest (less than 0.5% overall) node count label combinations. Therefore, this plot breaks down the "others" slice of the pie chart above. Values under 0.01% will be grouped into "others" to get a cleaner plot.

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Nodes per label combination (less than 0.5% overall)

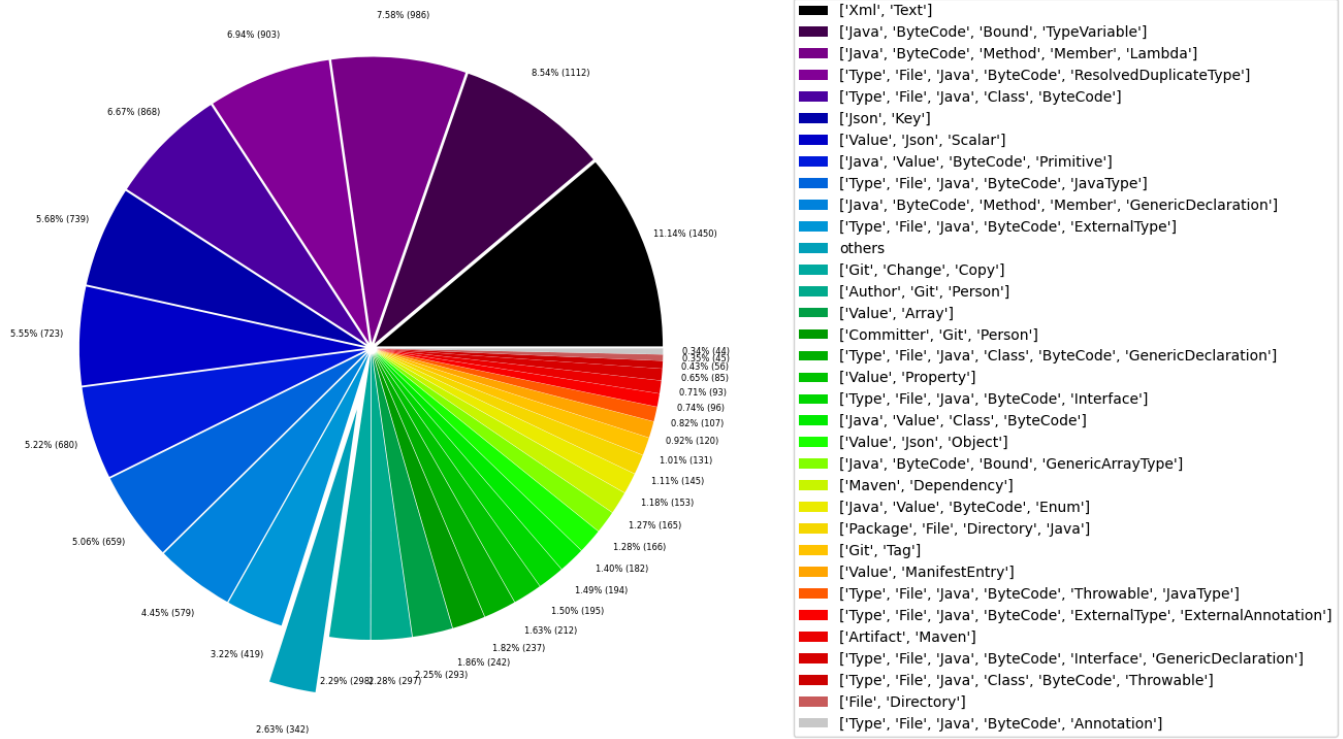


Table 1c - Highest node count by single label

Lists the 40 labels with the highest number of nodes. Doesn't sum up to the total number of nodes or 100% because one node can have multiple labels. Helps to identify commonly used labels.

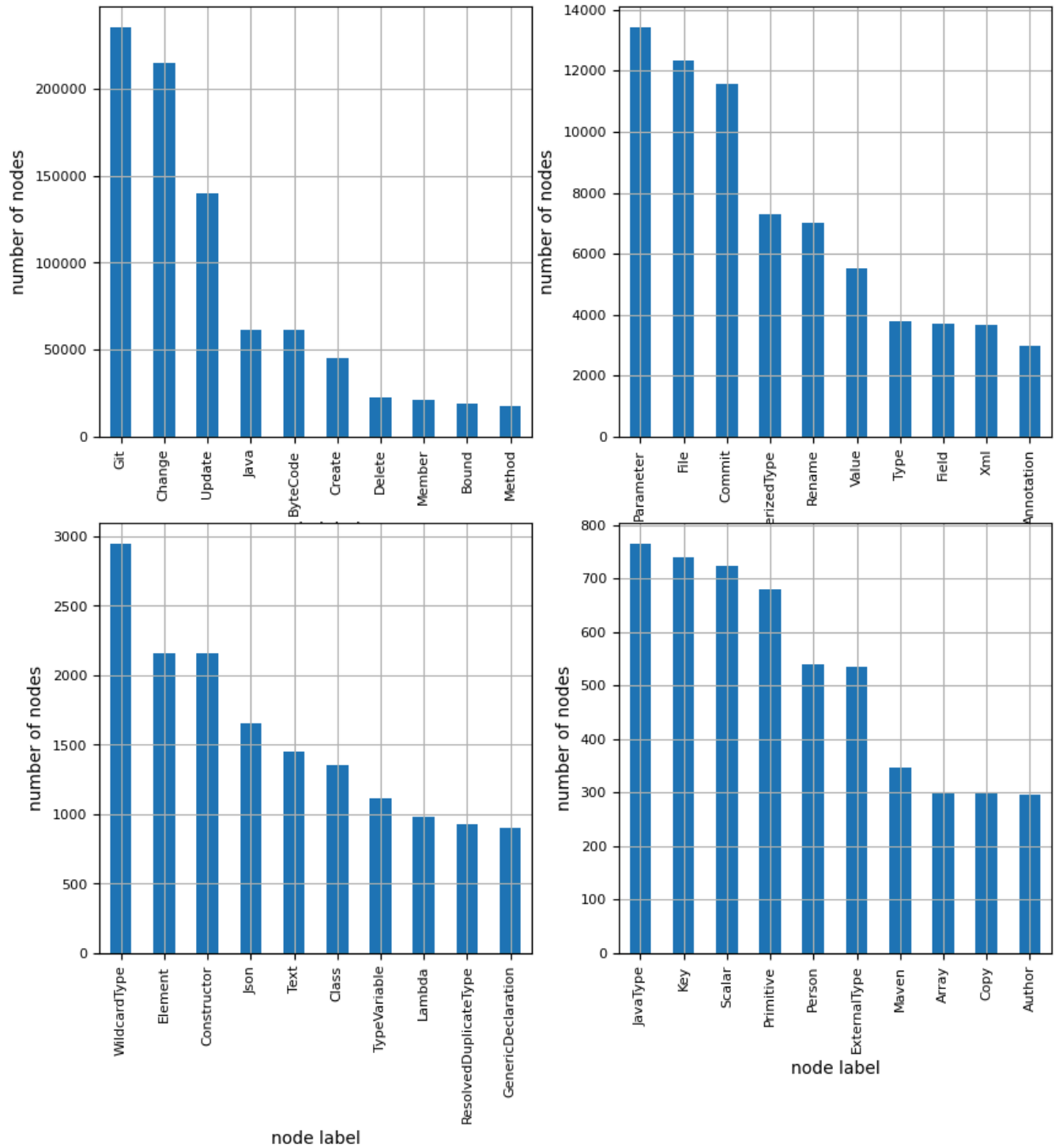
	nodeLabel	nodesWithThatLabel	nodesWithThatLabelPercent
0	Git	235613	77.652941
1	Change	215048	70.875162
2	Update	139612	46.013091
3	Java	61448	20.251930
4	ByteCode	61253	20.187662
5	Create	45325	14.938138
6	Delete	22795	7.512738
7	Member	21196	6.985742
8	Bound	18848	6.211893
9	Method	17504	5.768939
10	Parameter	13445	4.431181
11	File	12359	4.073259
12	Commit	11562	3.810585
13	ParameterizedType	7316	2.411195
14	Rename	7018	2.312981
15	Value	5518	1.818613
16	Type	3782	1.246465
17	Field	3692	1.216803
18	Xml	3675	1.211200
19	Annotation	2999	0.988405
20	WildcardType	2950	0.972256
21	Element	2162	0.712548
22	Constructor	2161	0.712219
23	Json	1652	0.544463
24	Text	1450	0.477889
25	Class	1355	0.446579
26	TypeVariable	1112	0.366491
27	Lambda	986	0.324964
28	ResolvedDuplicateType	925	0.304860
29	GenericDeclaration	905	0.298268
30	JavaType	766	0.252457
31	Key	739	0.243558
32	Scalar	723	0.238285
33	Primitive	680	0.224113
34	Person	539	0.177643
35	ExternalType	536	0.176654
36	Maven	346	0.114034
37	Array	299	0.098544
38	Copy	298	0.098214
39	Author	297	0.097885

Chart 1c - Highest node count by label

Shows the 40 labels with the highest number of nodes.

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Node count by label



Relationship Types

Table 2a - Highest relationship count by type

Lists the 30 relationship types with the highest number of occurrences. The whole table can be found in the CSV report `Relationship_type_count`.

Total number of relationships: 943511

	relationshipType	nodesWithThatRelationshipType	nodesWithThatRelationshipTypePercent
0	CONTAINS_CHANGE	215048	22.792315
1	MODIFIES	215048	22.792315
2	UPDATES	139612	14.797072
3	CREATES	52641	5.579267
4	INVOKES	37360	3.959678
5	DELETES	29813	3.159794
6	COMMITTED	23124	2.450846
7	DEPENDS_ON	22888	2.425833
8	OF_TYPE	22197	2.352596
9	DECLARES	21661	2.295787
10	OF_RAW_TYPE	17460	1.850535
11	HAS	14595	1.546882
12	HAS_PARENT	14144	1.499082
13	RETURNS	13071	1.385357
14	HAS_COMMIT	11562	1.225423
15	READS	9669	1.024789
16	HAS_ACTUAL_TYPE_ARGUMENT	8445	0.895061
17	HAS_FILE	8331	0.882979
18	RENAMES	7018	0.743818
19	OF_GENERIC_TYPE	6036	0.639738
20	RESOLVES_TO	4240	0.449385
21	SIMILAR	4130	0.437727
22	WRITES	4042	0.428400
23	CONTAINS	4005	0.424478
24	HAS_NEW_NAME	3662	0.388125
25	RETURNS_GENERIC	3616	0.383249
26	CHANGED_TOGETHER_WITH	3069	0.325274
27	ANNOTATED_BY	2943	0.311920
28	REQUIRES	2267	0.240273
29	HAS_FIRST_CHILD	2162	0.229144

Chart 2a - Highest relationship count by type

Values under 0.5% will be grouped into "others" to get a cleaner plot. The group "others" is then broken down in the second chart.

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Relationship types (more than 0.5% overall)

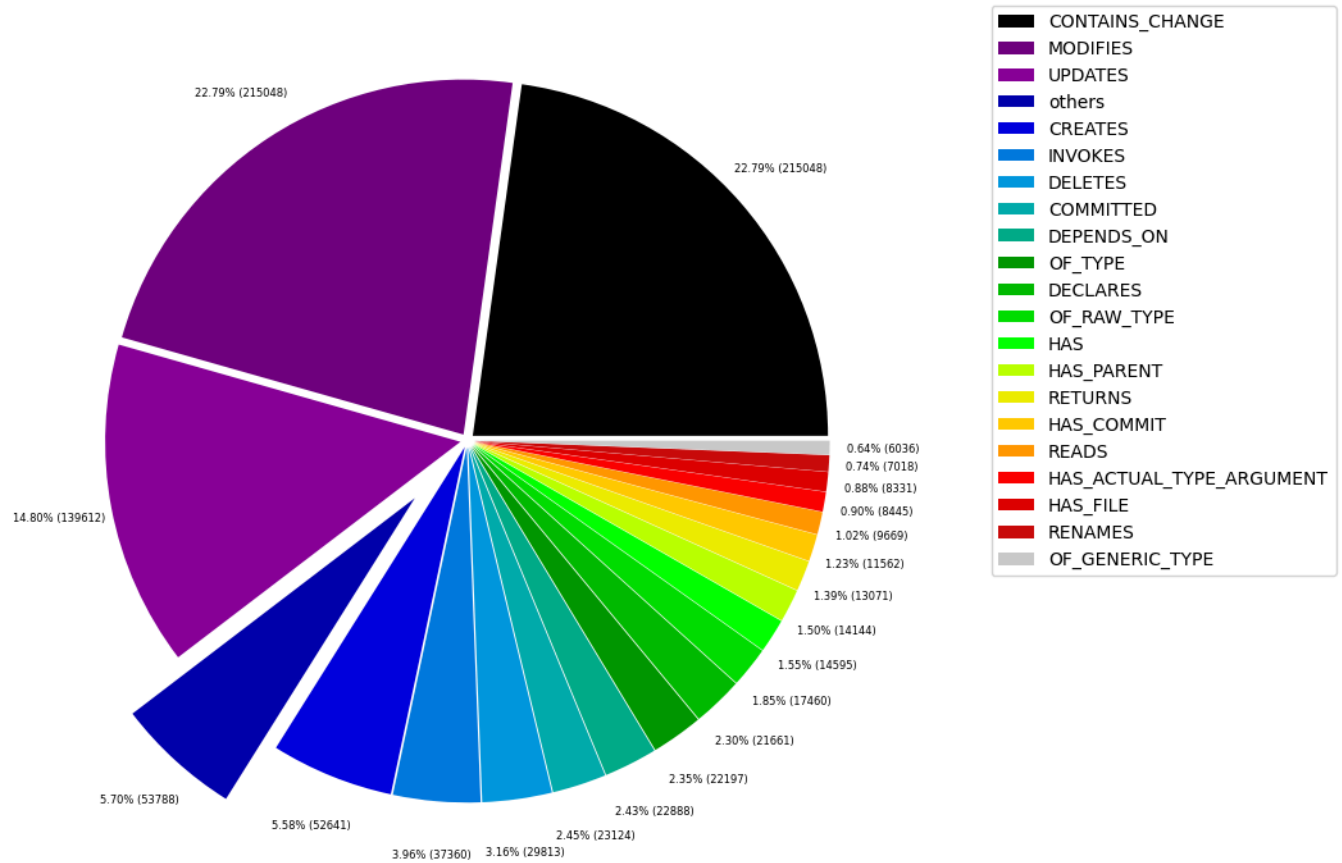


Table 2b - Lowest relationship count by type

Lists the 30 relationships type with the lowest number of occurrences up to 0.5% of the total node count. This is essentially breaking down the "others" slice from the chart above.

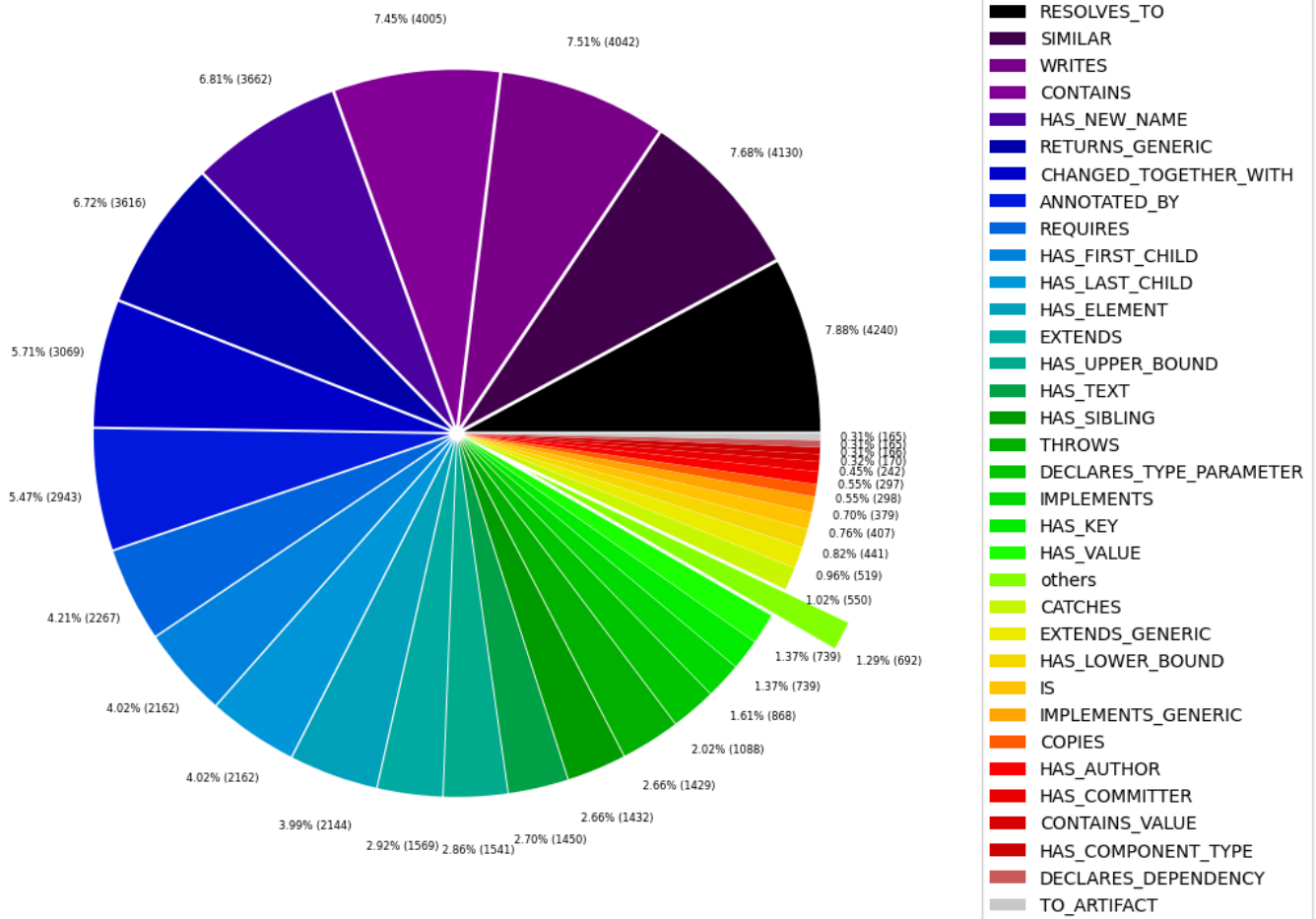
	relationshipType	nodesWithThatRelationshipType	nodesWithThatRelationshipTypePercent
0	HAS_PROPERTY	1	0.000106
1	HAS_BRANCH	1	0.000106
2	HAS_HEAD	2	0.000212
3	THROWS_GENERIC	5	0.000530
4	EXCLUDES	5	0.000530
5	DESCRIBES	9	0.000954
6	HAS_ROOT_ELEMENT	11	0.001166
7	HAS_GOAL	16	0.001696
8	HAS_EXECUTION	16	0.001696
9	OF_NAMESPACE	18	0.001908
10	HAS_ATTRIBUTE	18	0.001908
11	INCLUDES_CONCEPT	19	0.002014
12	USES_PLUGIN	21	0.002226
13	IS_ARTIFACT	21	0.002226
14	HAS_CONFIGURATION	21	0.002226
15	REQUIRES_TYPE_PARAMETER	24	0.002544
16	REQUIRES_CONCEPT	28	0.002968
17	DECLARES_NAMESPACE	36	0.003816
18	HAS_DEFAULT	39	0.004133
19	COPY_OF	119	0.012612
20	HAS_TAG	131	0.013884
21	ON_COMMIT	131	0.013884
22	DECLARES_DEPENDENCY	165	0.017488
23	TO_ARTIFACT	165	0.017488
24	HAS_COMPONENT_TYPE	166	0.017594
25	CONTAINS_VALUE	170	0.018018
26	HAS_COMMITTER	242	0.025649
27	HAS_AUTHOR	297	0.031478
28	COPIES	298	0.031584
29	IMPLEMENTS_GENERIC	379	0.040169

Chart 2b - Lowest relationship count by type

Shows the lowest (less than 0.5% overall) relationship types. This plot breaks down the "others" slice of the pie chart above. Values under 0.01% will be grouped into "others" to get a cleaner plot.

<Figure size 640x480 with 0 Axes>

Relationship types (less than 0.5% overall)



Node labels with their relationships

Table 3a - Highest relationship count by node labels and relationship type

Lists the 30 node labels and their relationship types with the highest number of occurrences.

	sourceLabels	relationType	targetLabels	numberOfRelationships	numberOfNodesWithSameLabelsAsSource	numbe
0	[Git, Commit]	CONTAINS_CHANGE	[Git, Update, Change]	139612	11562	
1	[Git, Update, Change]	MODIFIES	[File, Git]	139612	139612	
2	[Git, Update, Change]	UPDATES	[File, Git]	139612	139612	
3	[Git, Commit]	CONTAINS_CHANGE	[Git, Change, Create]	45325	11562	
4	[Git, Change, Create]	MODIFIES	[File, Git]	45325	45325	
5	[Git, Change, Create]	CREATES	[File, Git]	45325	45325	
6	[Git, Commit]	CONTAINS_CHANGE	[Git, Change, Delete]	22795	11562	
7	[Git, Change, Delete]	MODIFIES	[File, Git]	22795	22795	
8	[Git, Change, Delete]	DELETES	[File, Git]	22795	22795	
9	[Java, ByteCode, Method, Member]	INVOKES	[Java, ByteCode, Method, Member]	22655	13676	
10	[Git, Commit]	HAS_PARENT	[Git, Commit]	14135	11562	
11	[Repository, File, Git]	HAS_COMMIT	[Git, Commit]	11562	1	
12	[Committer, Git, Person]	COMMITTED	[Git, Commit]	11562	242	
13	[Author, Git, Person]	COMMITTED	[Git, Commit]	11562	297	
14	[Java, ByteCode, Method, Member]	READS	[Java, ByteCode, Member, Field]	8745	13676	
15	[Java, ByteCode, Method, Member]	HAS	[Java, ByteCode, Parameter]	8558	13676	
16	[Repository, File, Git]	HAS_FILE	[File, Git]	8331	1	
17	[Git, Commit]	CONTAINS_CHANGE	[Git, Change, Rename]	7018	11562	
18	[Git, Change, Rename]	CREATES	[File, Git]	7018	7018	
19	[Git, Change, Rename]	DELETES	[File, Git]	7018	7018	
20	[Git, Change, Rename]	MODIFIES	[File, Git]	7018	7018	
21	[Git, Change, Rename]	RENAMES	[File, Git]	7018	7018	
22	[Java, ByteCode, Parameter]	OF_TYPE	[Type, File, Java, ByteCode, JavaType]	6234	13445	
23	[File, Git]	HAS_NEW_NAME	[File, Git]	3662	8331	
24	[Java, ByteCode, Bound]	OF_RAW_TYPE	[Type, File, Java, ByteCode, JavaType]	3488	7304	
25	[Java, ByteCode, ParameterizedType, Bound]	OF_RAW_TYPE	[Type, File, Java, ByteCode, JavaType]	3144	7316	
26	[Java, ByteCode, ParameterizedType, Bound]	HAS_ACTUAL_TYPE_ARGUMENT	[Java, ByteCode, Bound, WildcardType]	2950	7316	
27	[Java, ByteCode, Parameter]	OF_GENERIC_TYPE	[Java, ByteCode, ParameterizedType, Bound]	2709	13445	
28	[Java, ByteCode, Constructor, Method, Member]	WRITES	[Java, ByteCode, Member, Field]	2577	2157	
29	[Java, ByteCode, ParameterizedType, Bound]	HAS_ACTUAL_TYPE_ARGUMENT	[Java, ByteCode, Bound, TypeVariable]	2477	7316	

Graph Density

```
total_number_of_nodes (vertices): 303418
total_number_of_relationships (edges): 943511
-> total directed graph density: 1.0248627689595746e-05
-> total directed graph density in percent: 0.0010248627689595745
```